

The Science of Seeing

 **Green Level: I Want More Support**

Name: _____ Date: _____ Period: _____

How Your Eyes Work

Your eyes collect light, but your **brain** creates the picture you see! Light travels through your eye and gets turned into electrical signals that your brain interprets.

Activity 1: Label the Eye

Use the **Word Bank** to label the parts of the eye.

Word Bank: Cornea • Pupil • Lens • Retina • Optic Nerve

[Eye Diagram Here - Teacher will provide]

1. _____ (clear front part, bends light)
2. _____ (black circle, controls light amount)
3. _____ (focuses light onto back of eye)
4. _____ (light-sensitive layer with rods and cones)
5. _____ (carries signals to brain - your blind spot!)

Activity 2: The Blind Spot Experiment

Follow these steps:

1. Close your **LEFT** eye
2. Look at the + with your **RIGHT** eye
3. Slowly move the paper closer to your face
4. The dot should disappear!



Why does this happen? Circle the correct answer:

- A) The dot moves off the paper
- B) Your optic nerve has no light receptors (blind spot)
- C) You blinked

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Two Types of Illusions

Eye-based illusions: Trick the cells in your eye (like afterimages that tire out your cones)

Brain-based illusions: Trick how your brain interprets what it sees (like size illusions)

Match the Illusion!

Draw a line from each illusion to where it happens:

Color Afterimage

Size Illusion

Blind Spot

EYE (retina)

BRAIN (visual cortex)

Mini Museum Connection

The **stereoscope** in the Mini Museum tricks your brain! It shows each eye a slightly different picture. Your brain combines them into one 3D image. This is called **binocular vision**.

Visit: minimuseumproject.com/exhibitions/seeing-is-deceiving

What I Learned

Fill in the blanks:

My eyes collect _____, but my _____ creates the picture I see.

The blind spot is where the _____ connects to my eye.

Optical illusions work by tricking either my _____ or my _____.

Share with Your Family!

Try the blind spot experiment with your family! Can they find their blind spot too?

Fun fact to share: Your blind spot is about the size of 17 full moons in the sky—but you never notice it because your brain fills it in!